

D And F Block**JEE Main 2020 Chapterwise****Questions with Answer Keys****Chemistry****Q1 JEE Main 2020 - 2 September (Evening)**

The oxidation states of transition metal atoms in $\text{K}_2\text{Cr}_2\text{O}_7$, KMnO_4 and K_2FeO_4 , respectively, are x , y and z . The sum of x , y and z is

Q2 JEE Main 2020 - 3 September (Evening)

The incorrect statement is

- (A) Manganate and permanganate ions are paramagnetic
- (B) Manganate and permanganate ions are tetrahedral
- (C) Manganate ion is green in colour and permanganate ion is purple in colour
- (D) In manganate and permanganate ions, the π -bonding takes place by overlap of p-orbitals of oxygen and d-orbitals of manganese

Q3 JEE Main 2020 - 4 September (Evening)

The incorrect statement(s) among (a) - (c) is (are)

- (a) W(VI) is more stable than Cr(VI) .
- (b) In the presence of HCl , permanganate titrations provide satisfactory results.
- (c) Some lanthanoid oxides can be used as phosphors.

- (A) (a) and (b) only
- (B) (a) only
- (C) (b) and (c) only
- (D) (b) only

Q4 JEE Main 2020 - 5 September (Morning)

The correct electronic configuration and spinonly magnetic moment (BM) of Gd^{3+} ($Z = 64$) respectively, are

- (A) $[\text{Xe}]5f^7$ and 8.9
- (B) $[\text{Xe}]4f^7$ and 7.9
- (C) $[\text{Xe}]5f^7$ and 7.9
- (D) $[\text{Xe}]4f^7$ and 8.9

Q5 JEE Main 2020 - 6 September (Morning)

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D And F Block

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

The lanthanoid that does NOT show + 4 oxidation state is

- (A) Tb
- (B) Dy
- (C) Ce
- (D) Eu

Q6 JEE Main 2020 - 6 September (Evening)

Mischmetal is an alloy consisting mainly of

- (A) lanthanoid and actinoid metals
- (B) actinoid and transition metals
- (C) lanthanoid metals
- (D) actinoid metals

Q7 JEE Main 2020 - 9 January (Morning)

The electronic configurations of bivalent europium and trivalent cerium are :

(atomic number : $Xe = 54$, $Ce = 58$, $Eu = 63$)

- (A) $[Xe] 4f^7$, $[Xe] 4f^1$
- (B) $[Xe] 4f^7 6s^2$, $[Xe] 4f^1$
- (C) $[Xe] 4f^7 6s^2$, $[Xe] 4f^1 5d^1 6s^2$
- (D) $[Xe] 4f^7$, $[Xe] 4f^1 5d^1 6s^2$

Q8 JEE Main 2020 - 9 January (Evening)

The sum of the total number of bonds between chromium and oxygen atoms in chromate and dichromate ions is

D And F Block

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Answer Key

Q1 (19)

Q2 (A)

Q3 (D)

Q4 (B)

Q5 (D)

Q6 (C)

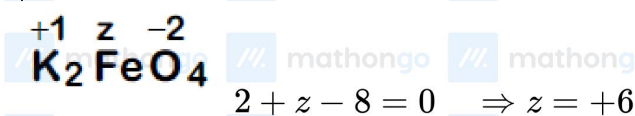
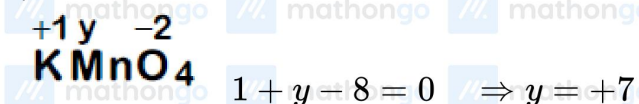
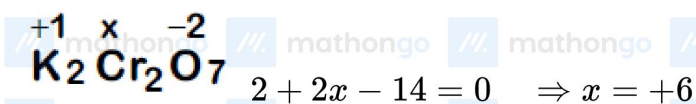
Q7 (A)

Q8 (12)

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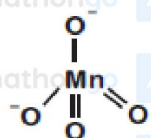
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Q1

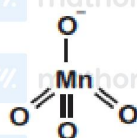


$$x + y + z = 6 + 7 + 6 = 19$$

Q2



Manganate ion (MnO_4^{2-})
Tetrahedral
Green colour
 π - $d\pi$ bonding
between oxygen and
manganese
diamagnetic

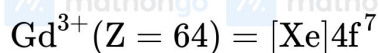


Permanganate ion (MnO_4^-)
Purple colour
 π - $d\pi$ bonding
between oxygen and
manganese
paramagnetic

Q3

- W(VI) is more stable than Cr(VI)
- Permanganate titrations in presence of HCl are unsatisfactory as HCl is oxidised to Cl_2
- Lanthanoid oxides are used as phosphors.

Q4



$$\mu = \sqrt{n(n+2)} = \sqrt{7(7+2)} = 7.9\text{B.M}$$

Q5

Europium (Eu)

Atomic No $\rightarrow 63$

Electronic configuration $\rightarrow [\text{Xe}] 4f^7 6s^2$

Can show only +2 and +3 oxidation state.

Q6

Misch metal is an alloy consisting mainly of lanthanoid metals.

Q7

$\text{Eu}^{2+} : [\text{Xe}] 4f^7$

$\text{Ce}^{3+} : [\text{Xe}] 4f^1$

Q8

$\Rightarrow$ Total number of Cr and O bonds is 12

